NAME: ______________________ **PERIOD:** ________

Centered or Close?

AP Statistics · Unit 3 · Topic 3.1

[STAR] THE PROBLEM

Suppose a research team tests two rules for estimating a support proportion in a validation population where the true proportion is $p = 0.40$. For a random sample of 25 people, let X be the number who support the proposal.

Rule A reports $\widehat{p}_A = X/25$. Rule B reports $\widehat{p}_B = (X+5)/35$. Exact summaries of the two sampling distributions are shown.

Lena: "A is unbiased, so it is always the more trustworthy estimator."

Omar: "B varies less and is more likely to land within 0.06 of the truth, so its bias does not matter."

Sampling distributions when $p = 0.40$

Rule	Mean	SD	Within 0.06
A	0.400	0.098	0.459
B	0.429	0.070	0.659

FIRST TAKE — one sentence before you work the parts

Which rule looks more useful for getting close to 0.40? Use one entry in the table.

[BOOK] BIAS AND VARIABILITY ANSWER DIFFERENT QUESTIONS (keep on the page)

A **point estimator** is a statistic used to estimate a population parameter; its observed value is a **point estimate**. An estimator is **unbiased** when the mean of its sampling distribution equals the parameter. **Bias** is long-run center minus the parameter; **variability** is sampling-distribution spread. Unbiased does not mean exact in one sample, and smaller spread does not erase bias. Judge the tradeoff against the stated goal.

Work (a)–(c).

DOK 1 **(a)** If one sample has $X = 9$ supporters, calculate the point estimate from Rule A and from Rule B. State the population parameter both rules estimate.

DOK 2 **(b)** Use the table to classify each rule as biased or unbiased. For any biased rule, give the direction and amount of bias; then compare the rules' sampling variability.

DOK 3 **(c)** In 3–4 sentences, choose a rule for getting within 0.06 of p and use the table to defend it. Explain what Lena and Omar each overlook about bias and variability.

Frame: For the within-0.06 goal, I would use Rule ____ because ______________________________.

Frame: That choice is not always better: its ______________ could make a reader believe ______________________________.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored E / P / I.